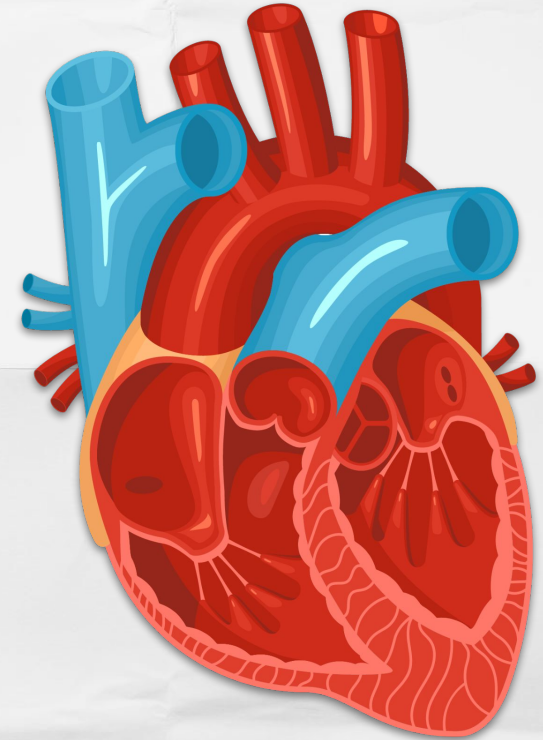
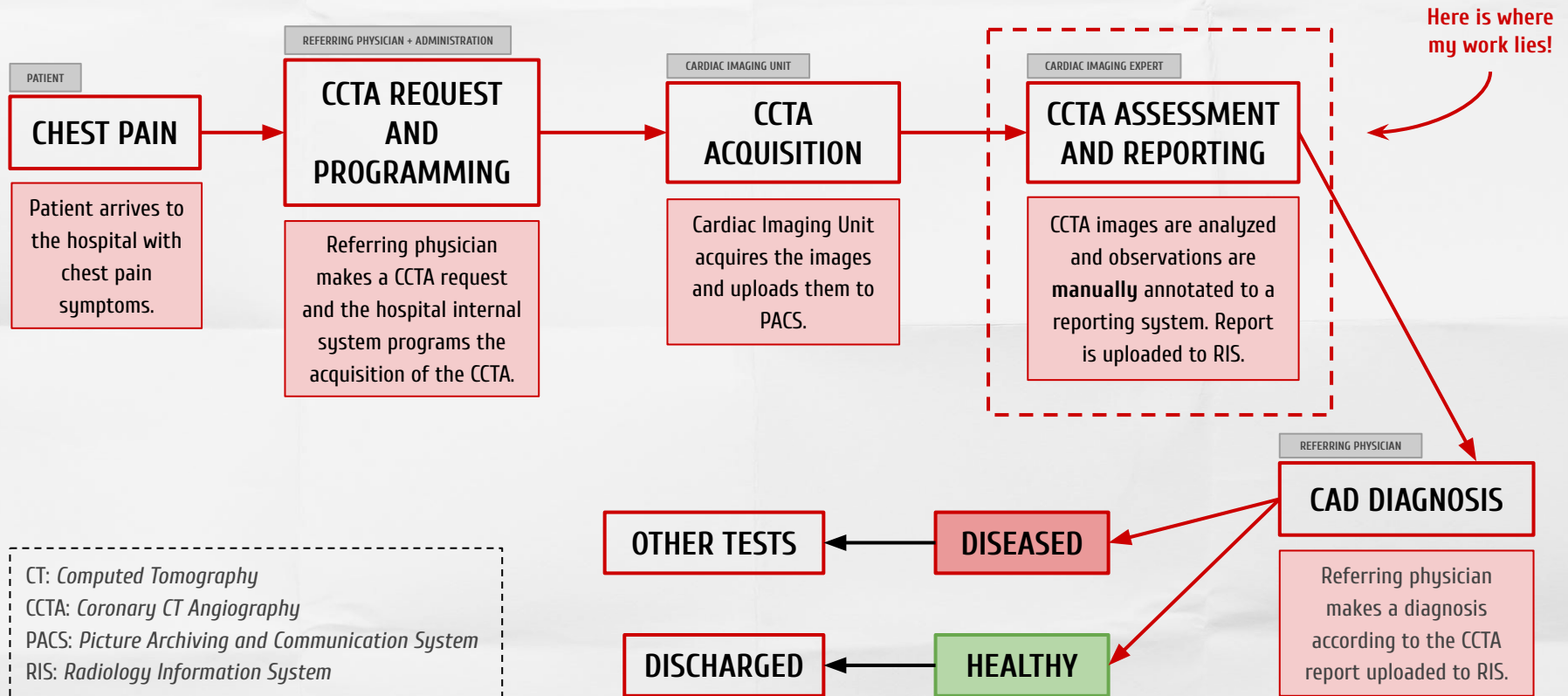


Automated Geometric Stenosis Quantification and CAD-RADS Support for Coronary Artery Disease Assessment and Patient Prioritization

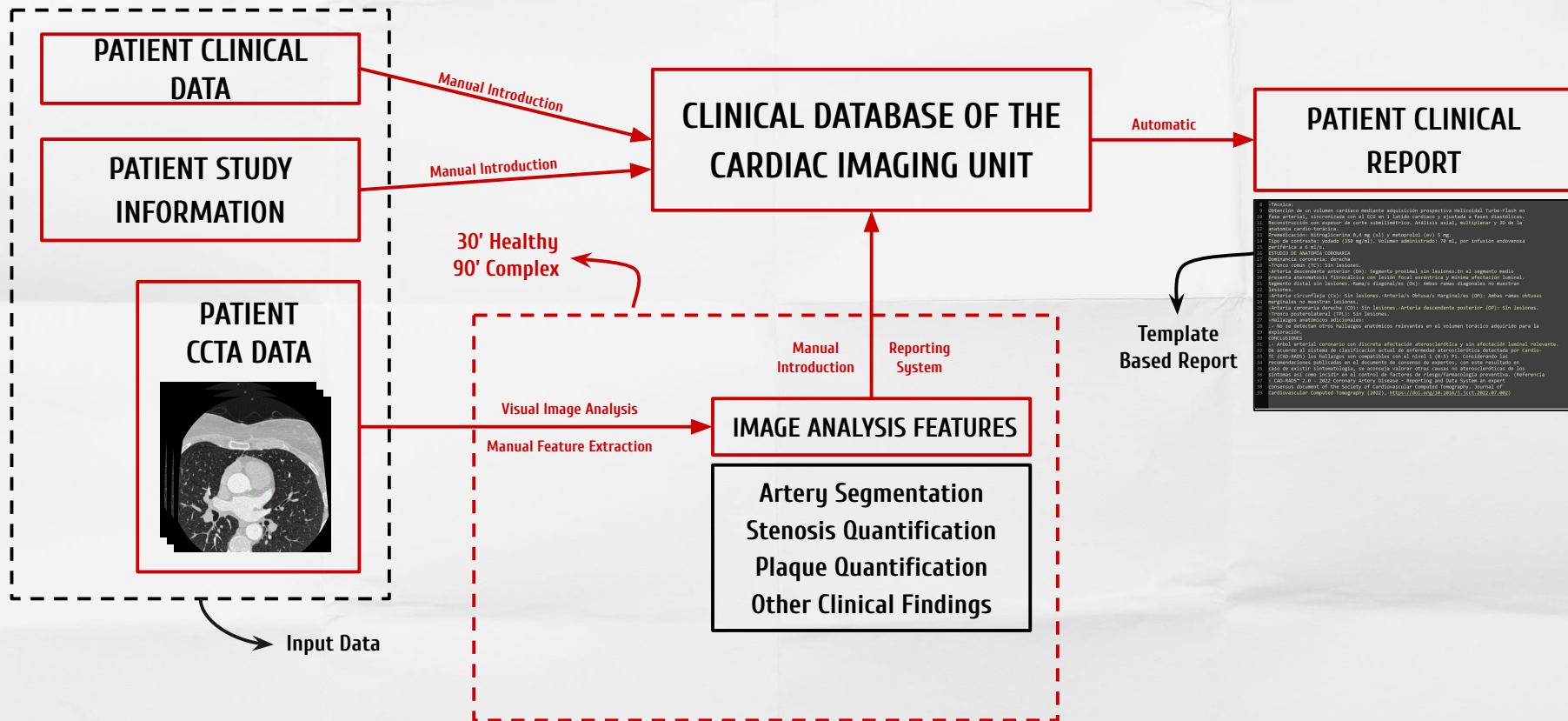
Adrià Cortés Cugat
Mathematical Engineering in Data Science



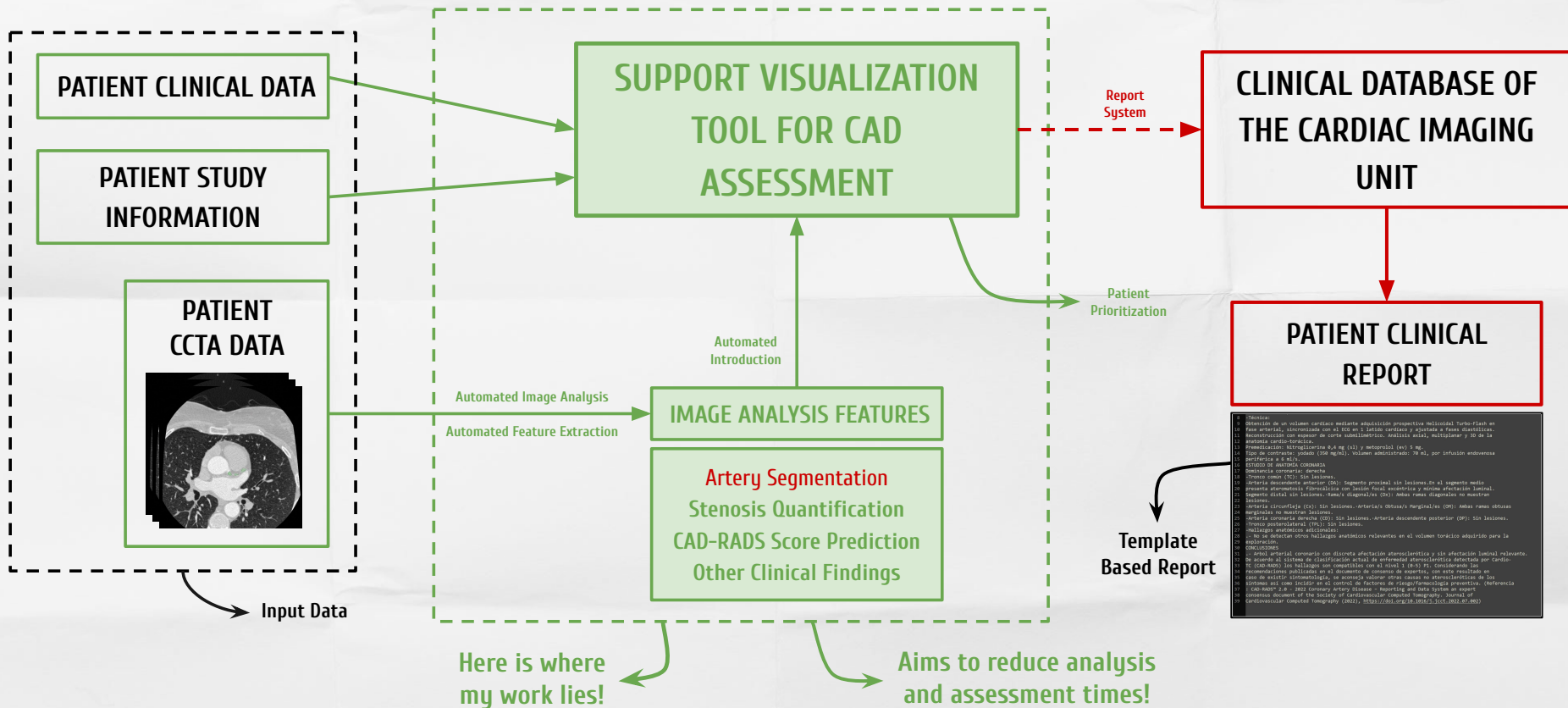
CURRENT CLINICAL WORKFLOW: PATIENT CAD DIAGNOSIS



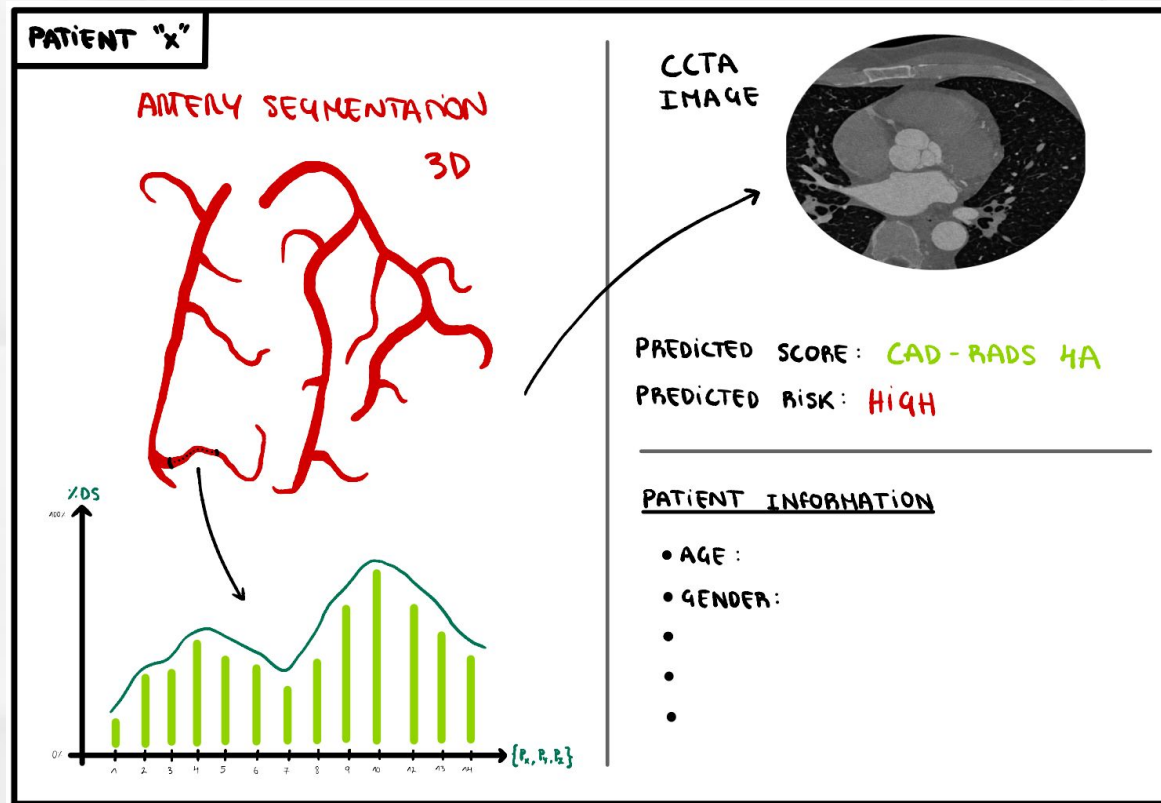
CURRENT CLINICAL WORKFLOW: IMAGE ANALYSIS AND DIAGNOSIS



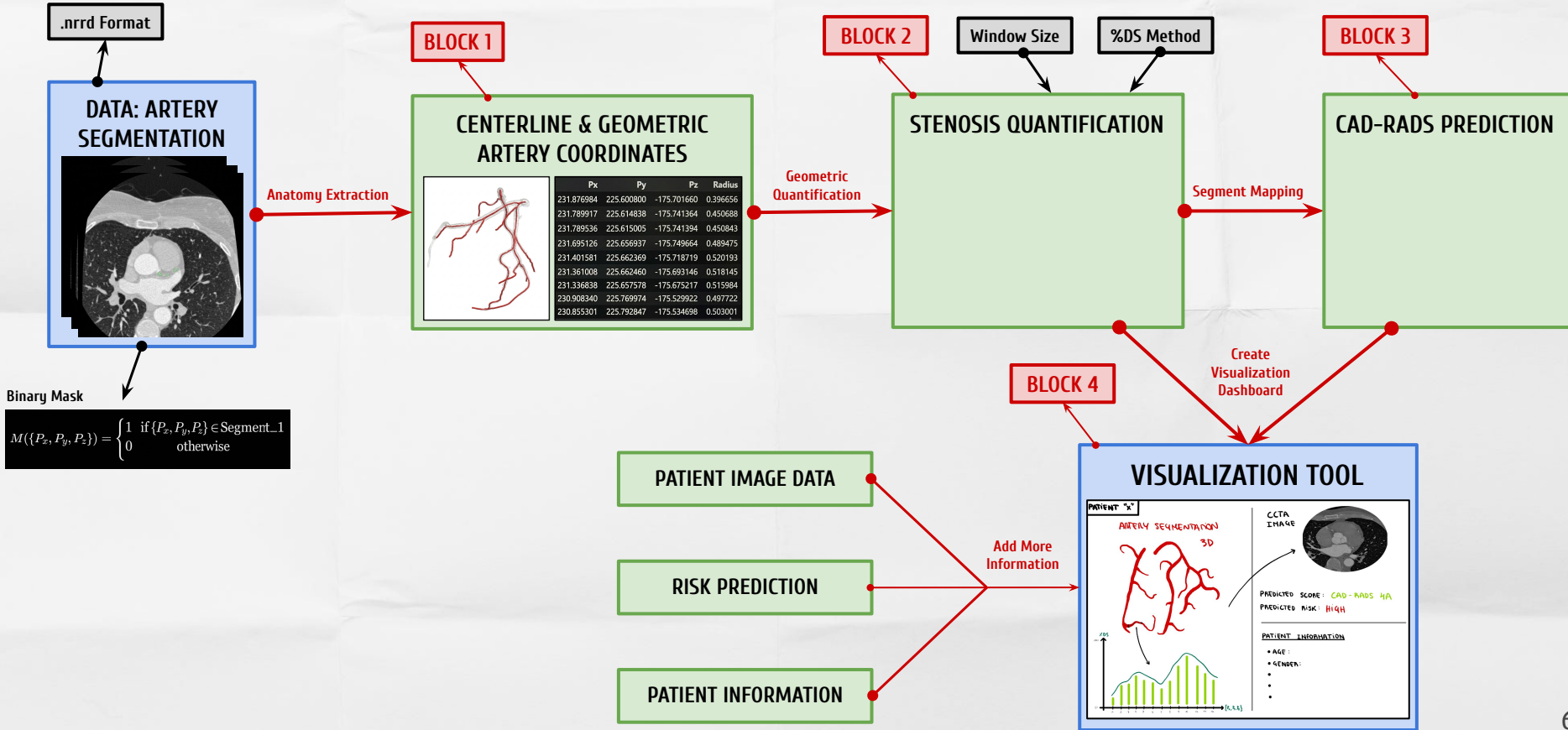
PROPOSED CLINICAL WORKFLOW: AUTOMATED SUPPORT FOR CAD ASSESSMENT

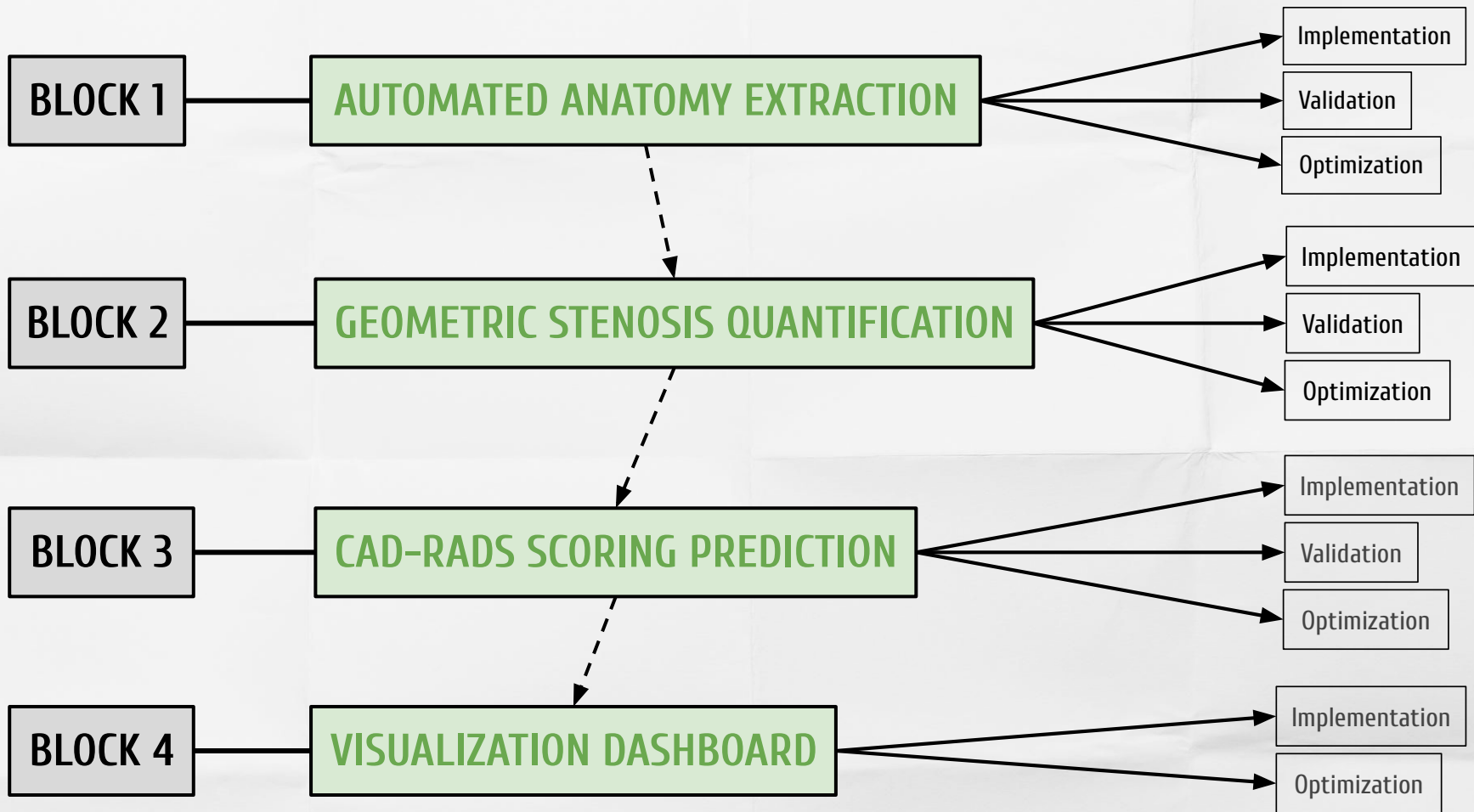


PROROTYPE FINAL SUPPORT VISUALIZATION TOOL FOR CAD ASSESSMENT



MASTER ARCHITECTURE: E2E AUTOMATED PIPELINE





FINAL DEGREE PROJECT DASHBOARD

B1	AUTOMATED ANATOMY EXTRACTION							
Image Loading	RCA/LCA Separation	Skeletonization	Endpoint Detection	Mesh Creation and Smoothing	Centerlines Extraction	Build Dataframe	VALIDATION	OPTIMIZATION

B2	GEOMETRIC STENOSIS QUANTIFICATION					
Area Computation		Reference Value Computation	Stenosis % Computation	Data Aggregation	VALIDATION	OPTIMIZATION

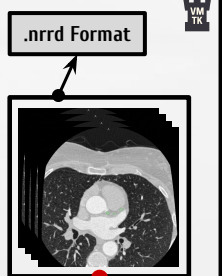
B3	CAD-RADS SCORING PREDICTION					
++++	++++			+++++	VALIDATION	OPTIMIZATION

B4	VISUALIZATION DASHBOARD				
3D Artery Mesh	Stenosis Visualizations	CAD-RADS Visualizations	Patient Information Visualization	VALIDATION	OPTIMIZATION

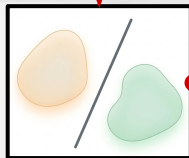
BLOCK 1

AUTOMATED ANATOMY EXTRACTION

Artery Seed Points



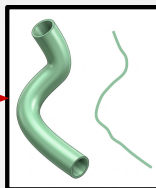
Load Binary Mask
Split RCA/LCA Using Mass Center



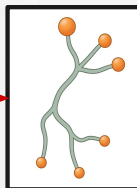
Input: Artery Segmentation Data

For Each Component

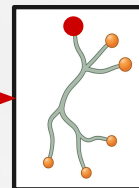
Voxel Skeletonization



Endpoint Detection



Ostium Identification

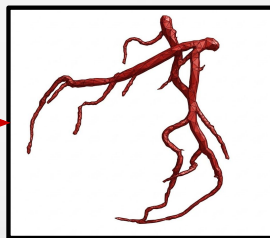


Point Coordinate Conversion

Voxel $\rightarrow \{Px, Py, Pz\}$

Artery 3D Mesh Smoothed

Artery Mech Creation & Smoothing



Centerline Extraction

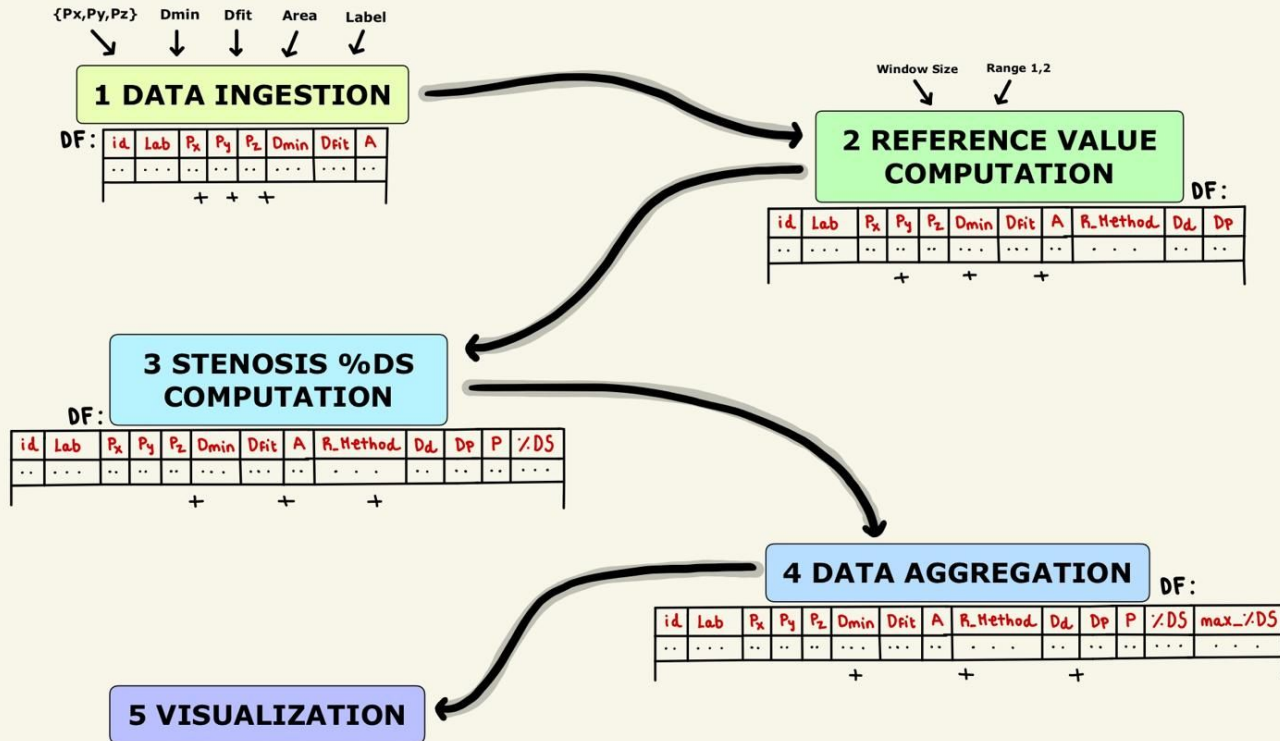


Artery Geometry Data Frame

0	Sample_ID	Artery_Type	Px	Py	Pz	Radius
1	1	RCA	231.876984	225.600800	-175.701660	0.396656
2	1	RCA	231.789917	225.614838	-175.741364	0.450688
3	1	RCA	231.789536	225.615005	-175.741394	0.450843
4	1	RCA	231.695126	225.656937	-175.749664	0.489475
5	1	RCA	231.401581	225.662369	-175.718719	0.520193
6	1	RCA	231.361008	225.662460	-175.693146	0.518145
7	1	RCA	231.336838	225.657578	-175.675217	0.515984
8	1	RCA	230.908340	225.769974	-175.529922	0.497722
9	1	RCA	230.855301	225.792847	-175.534698	0.503001

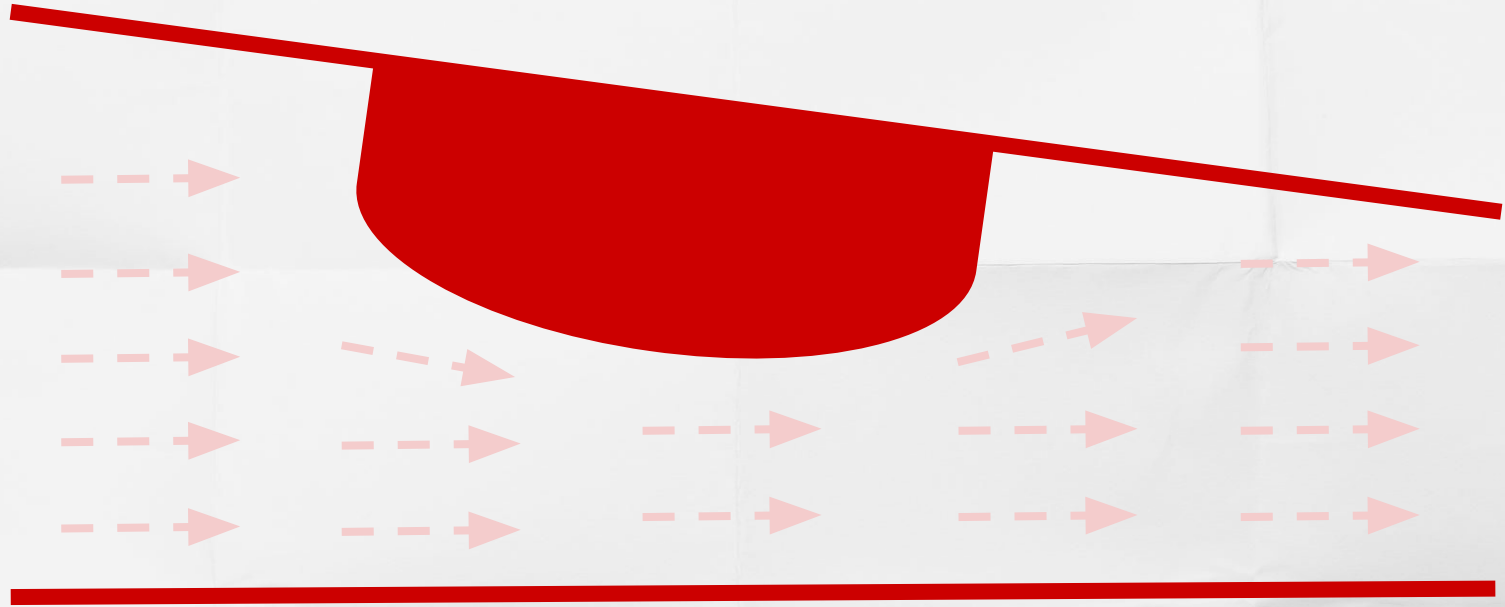
BLOCK 2

GEOMETRIC STENOSIS QUANTIFICATION



BLOCK 2

GEOMETRIC STENOSIS QUANTIFICATION



BLOCK 2

GEOMETRIC STENOSIS QUANTIFICATION

Px	Py	Pz	Radius
231.876984	225.600800	-175.701660	0.396656
231.789917	225.614838	-175.741364	0.450688
231.789536	225.615005	-175.741394	0.450843
231.695126	225.656937	-175.749664	0.489475
231.401581	225.662369	-175.718719	0.520193
231.361008	225.662460	-175.693146	0.518145
231.336838	225.657578	-175.675217	0.515984
230.908340	225.769974	-175.529922	0.497722
230.855301	225.792847	-175.534698	0.503001

BLOCK 3

CAD-RADS SCORING PREDICTION

BLOCK 4

VISUALIZATION DASHBOARD



Universitat
Pompeu Fabra
Barcelona



SANT PAU
Campus Salut
Barcelona



Hospital de
la Santa Creu i
Sant Pau

